

Quadratic deformations of the game exterior algebra

Codex

2026-06-17

[RESOLVED] Closure addendum (2026-08-09). The local coefficient analysis preserved in this note has now been superseded at the global boundary by `writeups/game_exterior_divisibility.tex`. Moews’s structure theorem makes multiplication by every power of two surjective on the full short-game group. If a torsion game t has order $n = 2^k$, choose $ny = t$ and $nz = x$; then $n^2y = 0$, so every additive grade-one realization satisfies

$$e_t^2 = (ne_y)^2 = 0, \quad e_te_x + e_xe_t = 0.$$

Coefficient faithfulness therefore forces $Q(t) = B(t, x) = 0$ in every coefficient characteristic. Thus no nonzero torsion table extends as an ambient-coherent game-native Clifford deformation. The root-incomplete local tables studied below remain valid as local algebra, but are not residual cases of the resolved global question.

Abstract

This note preserves an earlier attack on the then-open problem “quadratic deformation of the game exterior algebra.” The implementation has the exterior algebra $\Lambda_{\mathbb{Z}}M$ of a finitely generated game subgroup M , plus a checked integer-valued Clifford surface that accepts hand-supplied quadratic data only when the game relations are compatible. The new observation is that there are two different descent gates. A quadratic map must descend through the game relations; a Clifford quotient over a coefficient ring must also keep the coefficient ring from being accidentally collapsed by the two-sided relation ideal. Over $\mathbb{Z}$ both gates kill torsion squares. Over torsion coefficient rings the first gate can reopen, but the second gate rules out several tempting “universal” torsion lifts.

1 Live boundary

[SOURCE-PINNED] The implemented object is the following. Given games $g_1, \dots, g_n$, the module layer forms the subgroup

$$M = \langle g_1, \dots, g_n \rangle_{\mathbb{Z}}$$

of the normal-play game group, represented by generators and integer relation rows. `src/games/game_exterior/lambda.rs` builds $\Lambda_{\mathbb{Z}}M$ by starting from the free Grassmann algebra and quotienting by the exterior ideal generated by those relation rows. For example, the relation $2\star = 0$ propagates to $2(\star \wedge \uparrow) = 0$.

[SOURCE-PINNED] `src/games/game_exterior/clifford.rs` is deliberately smaller than the research problem. It builds an integer Clifford algebra from hand-supplied diagonal values $q_i = Q(e_i)$ and off-diagonal polar values $b_{ij} = B(e_i, e_j)$, then accepts the data only if every game relation is null and polar-radical. This is a checked engineering surface, not a source of game-native quadratic data.

[TESTED] The focused test suite was run during this pass:

`cargo test game_exterior.`

It passed all twelve filtered tests, including the tests that reject integer-valued torsion squares and accept the documented torsion vanishings.

2 Quadratic descent

Let R be a commutative coefficient ring. A quadratic datum on an abelian group M is a map $Q: M \rightarrow R$ with polar form

$$B(x, y) = Q(x + y) - Q(x) - Q(y)$$

such that B is biadditive and $Q(-x) = Q(x)$. Equivalently, after choosing generators, the Clifford relations are

$$e_i^2 = q_i, \quad e_i e_j + e_j e_i = b_{ij}.$$

If a relation row $r = \sum_i c_i e_i$ represents zero in M , the quadratic data descends through the relation only if

$$Q(r) = 0, \quad B(r, e_j) = 0 \quad \text{for every } j.$$

These are exactly the null and polar-radical checks implemented over $\mathbb{Z}$.

Proposition 1 (torsion over torsion-free targets). **[PROVED]** *Let A be a torsion-free abelian target and let $t \in M$ have finite order. For every relation-respecting quadratic datum $Q: M \rightarrow A$, one has*

$$Q(t) = 0, \quad B(t, x) = 0 \quad \text{for every } x \in M.$$

Proof. If $nt = 0$, then $0 = B(nt, x) = nB(t, x)$ for every x , so torsion-freeness gives $B(t, x) = 0$. Also $0 = Q(nt) = n^2 Q(t)$, hence $Q(t) = 0$. $\square$

Corollary 2 (integer game-Clifford torsion split). **[PROVED]** *Every integer-valued deformation of a mixed subgroup $M = T \oplus F$, where T is torsion and F is free, is blind to T : the torsion generators remain exterior/nilpotent and are polar-orthogonal to the free generators. All nonzero integer quadratic data factors through the free quotient.*

This explains the implemented examples. In $\langle \star, \uparrow \rangle$, the relation $2\star = 0$ forces $Q(\star) = 0$ and $B(\star, \uparrow) = 0$ over $\mathbb{Z}$, while $Q(\uparrow)$ can still be chosen freely.

3 Clifford-faithful descent is stronger

For a pure quadratic map, the relation $nt = 0$ imposes $n^2 Q(t) = 0$. For a Clifford algebra quotient, that is not the whole story. The relation vector ne_t is placed in a two-sided ideal. Multiplying it by generators can produce coefficient scalars, so a quotient may accidentally kill part of R .

Proposition 3 (the coefficient-faithfulness obstruction). **[PROVED]** *Let R be a commutative coefficient ring and suppose a Clifford-type quotient imposes the module relation $ne_t = 0$ while keeping the coefficient map $R \rightarrow \text{Cl}$ injective. Then*

$$nQ(t) = 0, \quad nB(t, x) = 0 \quad \text{for every } x \in M.$$

Proof. Since ne_t is zero in the quotient, so is $(ne_t)e_t = nQ(t)$. If the coefficient ring injects, this scalar was already zero in R . Similarly

$$(ne_t)e_x + e_x(ne_t) = nB(t, x)$$

is a scalar killed by the relation ideal, so coefficient-faithfulness forces it to be zero in R . $\square$

Remark 4. Over $R = \mathbb{Z}$, Proposition 3 gives the same visible answer as Proposition 1: torsion squares vanish. Over torsion rings it is strictly sharper. For a 2-torsion game t , the formal quadratic descent condition only asks $4Q(t) = 0$, but a faithful Clifford quotient asks $2Q(t) = 0$.

3.1 The $\mathbb{Z}/4$ Brown lift is not a faithful square quotient

Take $M = \langle \star \rangle \cong \mathbb{Z}/2$ and try $R = \mathbb{Z}/4$ with $Q(\star) = 1$. As a bare quadratic map, this passes $Q(2\star) = 4 = 0$ in $\mathbb{Z}/4$. But the Clifford relation row is $2e_\star = 0$, and

$$(2e_\star)e_\star = 2e_\star^2 = 2.$$

Thus the two-sided relation ideal kills the scalar 2. The quotient has silently collapsed the coefficient ring from $\mathbb{Z}/4$ toward characteristic 2. This is not a faithful $\mathbb{Z}/4$ -valued deformation.

[SOURCE-PINNED] This does not say that $\mathbb{Z}/4$ -valued quadratic refinements are invalid. They are exactly the Brown-invariant category implemented in `src/forms/char2/brown.rs`: maps $q: V \rightarrow \mathbb{Z}/4$ on an $\mathbb{F}_2$ -space with

$$q(x + y) = q(x) + q(y) + 2b(x, y).$$

That is a genuine quadratic-module target and a real mod-8 invariant. The point here is narrower: a Brown refinement is not automatically a faithful Clifford square relation $e_x^2 = q(x)$ after also imposing the game-module relation $2e_x = 0$. The Brown route belongs to the quadratic-module/Form wing; it does not by itself deform `GameExterior`'s algebra without changing the coefficient ring seen by the quotient.

3.2 The $\mathbb{F}_2$ central-square escape is real but small

With $R = \mathbb{F}_2$, the same one-generator subgroup can carry the nonzero datum $Q(\star) = 1$ without killing a coefficient scalar: $2 = 0$ already in R . The resulting algebra is a central-square characteristic-2 Clifford algebra, not a Grassmann algebra.

For a general subgroup M , the canonical mod-2 reduction gives $V = M/2M$. The tautological construction

$$R = \text{Sym}_{\mathbb{F}_2}(V), \quad Q(x) = \bar{x} \in V \subset R, \quad B = 0$$

is relation-compatible and coefficient-faithful after scalar extension to characteristic 2. It is also too formal to settle the open problem: it has zero polar form, all generators commute, and the coefficient ring is just the free polynomial record of the mod-2 game class. It records torsion instead of explaining it.

4 Natural positive sources that do not solve torsion

There is a useful family of positive examples. If

$$\phi: M \rightarrow A$$

is an additive game invariant into a commutative torsion-free ring, then

$$Q_\phi(x) = \phi(x)^2, \quad B_\phi(x, y) = 2\phi(x)\phi(y)$$

is a natural quadratic datum. It automatically respects game relations.

[SOURCE-PINNED] Two implemented game invariants supply instances of this pattern:

- The thermographic mean value is additive in `src/games/thermography.rs`; this pass ran `cargo test mean_value_is_additive`, which passed.
- Atomic weight is additive on all-small games in `src/games/atomic_weight.rs`. On subgroups where it lands in the integer-valued slice, it can be squared as above; this pass ran `cargo test atomic_weight_is_additive`, which passed.

For example, on the all-small subgroup generated by $\uparrow$ and $\star$, atomic weight gives

$$\text{aw}(\uparrow) = 1, \quad \text{aw}(\star) = 0.$$

Therefore $Q_{\text{aw}}(\uparrow) = 1$ while $Q_{\text{aw}}(\star) = 0$ and $B_{\text{aw}}(\star, \uparrow) = 0$. This is genuinely game-native on the free all-small direction, and it reproduces the mixed-subgroup split of Corollary 2. It does not produce nonzero torsion squares.

5 The nimber core is the one non-tautological torsion source

The finite nimber subgroup is different because it carries extra structure not available on arbitrary games: nim-addition, Turning-Corners multiplication, and trace. Existing Gold/Arf work uses this to build characteristic-2 quadratic forms

$$Q_a(x) = \text{Tr}(x x^{2^a})$$

on finite nimber fields. This is a real non-tautological torsion quadratic source, but it lives exactly on the field-like impartial core where the scalar story already applies. It does not extend the game exterior algebra over general partizan games; it confirms the boundary stated in `src/games/partizan.rs`: arbitrary games form an abelian group, not a ring.

6 Historical local boundary (superseded globally)

[RESOLVED] At the time of this pass, I could not find a non-tautological torsion square for arbitrary game subgroups. The alternatives then separated were:

1. Torsion-free coefficient targets. These are obstructed by Proposition 1; torsion games are forced into the exterior radical.
2. Torsion coefficient targets with only formal quadratic descent. These can pass $Q(nt) = 0$ while failing coefficient-faithful Clifford descent, as the $\mathbb{Z}/4$ example shows.
3. Characteristic-2 central-square targets. These can keep nonzero torsion squares, but the canonical construction through $M/2M$ is tautological and has zero polar form.
4. Nimber-field targets. These are genuinely game-native through Turning-Corners multiplication and trace, but only on the impartial field-like core.

5. Additive thermographic or atomic-weight sources. These are game-native and useful on free directions, but every torsion element maps to zero.

At that local stage, the remaining shape looked sharper than “find a metric.” A solving construction appeared to need a game-built coefficient target and a square operation that:

- survives the coefficient-faithfulness test of Proposition 3;
- is not merely the tautological polynomial ring on $M/2M$;
- does not factor through an additive invariant into a torsion-free ring;
- reaches beyond the nimber field core.

The later divisibility theorem closes all five commutative-scalar alternatives: adjoining the required short-game roots kills every torsion square and polar value before the coefficient characteristic matters. A game-native noncommutative or directed structure whose square remembers first-/second-player asymmetry would instead change the mathematical object and belongs to the separate outcome-semantics problem.

7 Implementation consequence

[PROVED] The current integer implementation is aligned with the obstruction: over $\mathbb{Z}$, null and polar-radical relation checks are enough to reject the torsion squares considered here. However, a future generalization of `GameClifford` to torsion coefficient rings should not reuse only those checks. It must also check the scalar intersection of the two-sided relation ideal, or at least the necessary conditions

$$nQ(t) = 0, \quad nB(t, x) = 0$$

for every visible torsion relation $nt = 0$. Otherwise a datum can look quadratic while the quotient silently changes the coefficient ring.